

# Make It Do Something — Submission & Explainer

**Student:** Abdul Sami Uthwal

**Selected Feature:** Live Lead Capture System (1 Feature)

**Live URL:** <https://abdulsamiuthwal-portfolio.vercel.app/#contact-page>

**Backend Provider:** Web3Forms API (Free Tier)

**Recipient Email:** abdulsamiuthwal@gmail.com

**Verification Status:** Real Test Submission Verified

## 1. Live Feature Overview & Proof of Functionality

For the Week 8 assignment, I selected and fully wired **exactly one dynamic feature** into my live portfolio website: an end-to-end **Lead Capture & Contact Engine**.

### Verified Test Submission Evidence:

A live test submission was performed directly on the production domain <https://abdulsamiuthwal-portfolio.vercel.app/#contact-page> with test payload (*Name: Test User, Email: test@example.com, Goals: Hello world test*). The serverless backend API processed the submission, and a formatted lead notification was successfully delivered in real-time to my Gmail inbox (abdulsamiuthwal@gmail.com).

## 2. Plain-Words Explainer

### A. What is a Backend?

In modern software development, a **frontend** is the interactive visual layer rendered inside the client's web browser (HTML layouts, CSS design tokens, form fields, and buttons). A **backend** is the server-side infrastructure running behind the scenes that executes business logic, manages database records, verifies security, and relays data to external services (like email providers or third-party APIs).

Without a backend, interactive form controls are purely aesthetic—clicking "Submit" has nowhere to send the data. The backend provides the underlying pipeline that translates user interactions into real-world outcomes.

### B. What My Feature Does

My portfolio features a single, production-wired **Lead Capture Engine**. When a hiring manager, client, or recruiter fills out the contact form on my live site, the feature receives their inputs (Name, Email,

Organization, Project Requirements). Instead of displaying a static alert or relying on broken `mailto:` links, it transmits the inquiry to a serverless backend API, which immediately formats and delivers a lead notification to my personal Gmail inbox so I can follow up directly.

### C. Step-by-Step Data Flow

1. **User Input (Client Layer):** The visitor fills out the contact fields on `https://abdulsamiuthwal-portfolio.vercel.app/#contact-page` and clicks "Send Message / Get Estimate".
2. **Client Dispatch (Transport Layer):** JavaScript intercepts the form submission event, packages the input fields into a structured JSON object alongside an API access key, and sends an asynchronous HTTP POST request to `https://api.web3forms.com/submit`.
3. **Backend Processing (Serverless Layer):** Web3Forms acts as the serverless backend. It validates the access key, checks for spam signatures, packages the JSON fields into a clean HTML email template, and relays it via SMTP.
4. **Inbox Delivery (Output Layer):** The formatted lead lands in my Gmail inbox (`abdulsamiuthwal@gmail.com`) within seconds, with the Reply-To header set to the visitor's email address for direct two-way communication.

## 3. Technical & Architecture Notes

- **Zero Cold-Start Delay:** By utilizing Web3Forms serverless API over standard custom Node/Python backend servers on free tiers (like Render spin-downs), submissions execute instantly without 50-second spin-up lags.
- **Free-Tier Compliance:** Operates under Web3Forms free tier (250 submissions/month), ensuring a cost-free, maintenance-free production deployment.
- **Rubric Adherence:** Focuses strictly on **exactly one feature** working end-to-end with real submission evidence, as specified in the assignment brief.